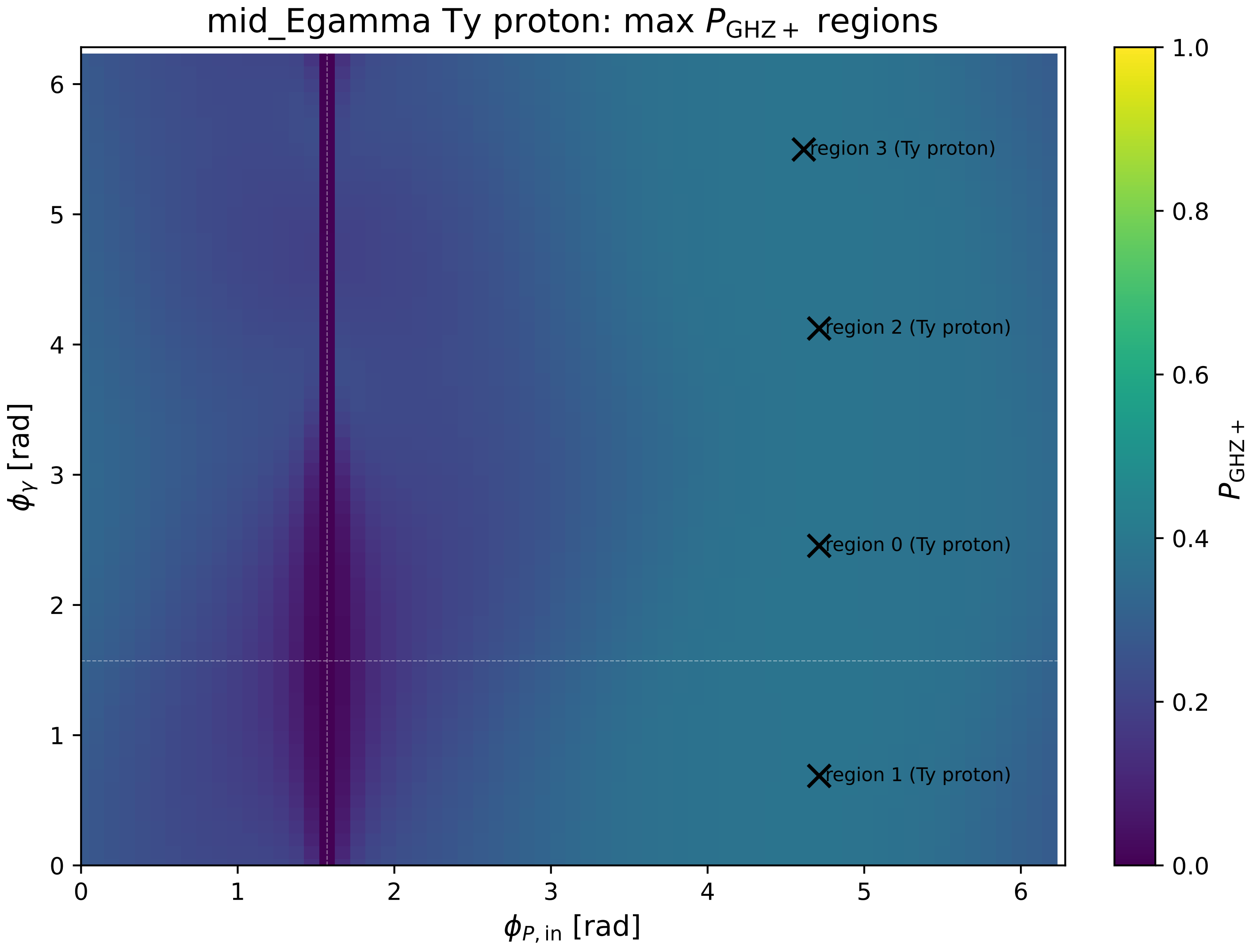
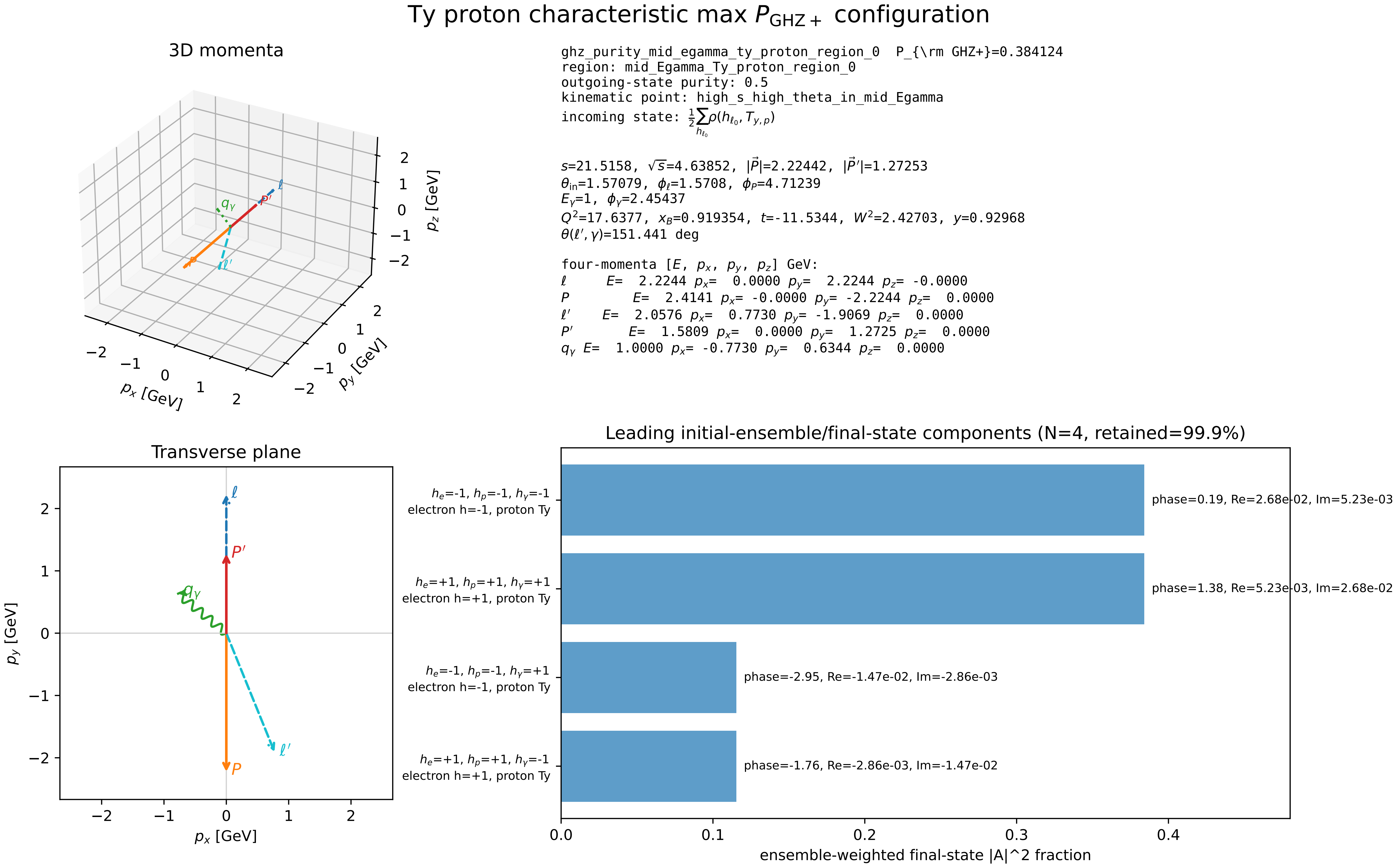
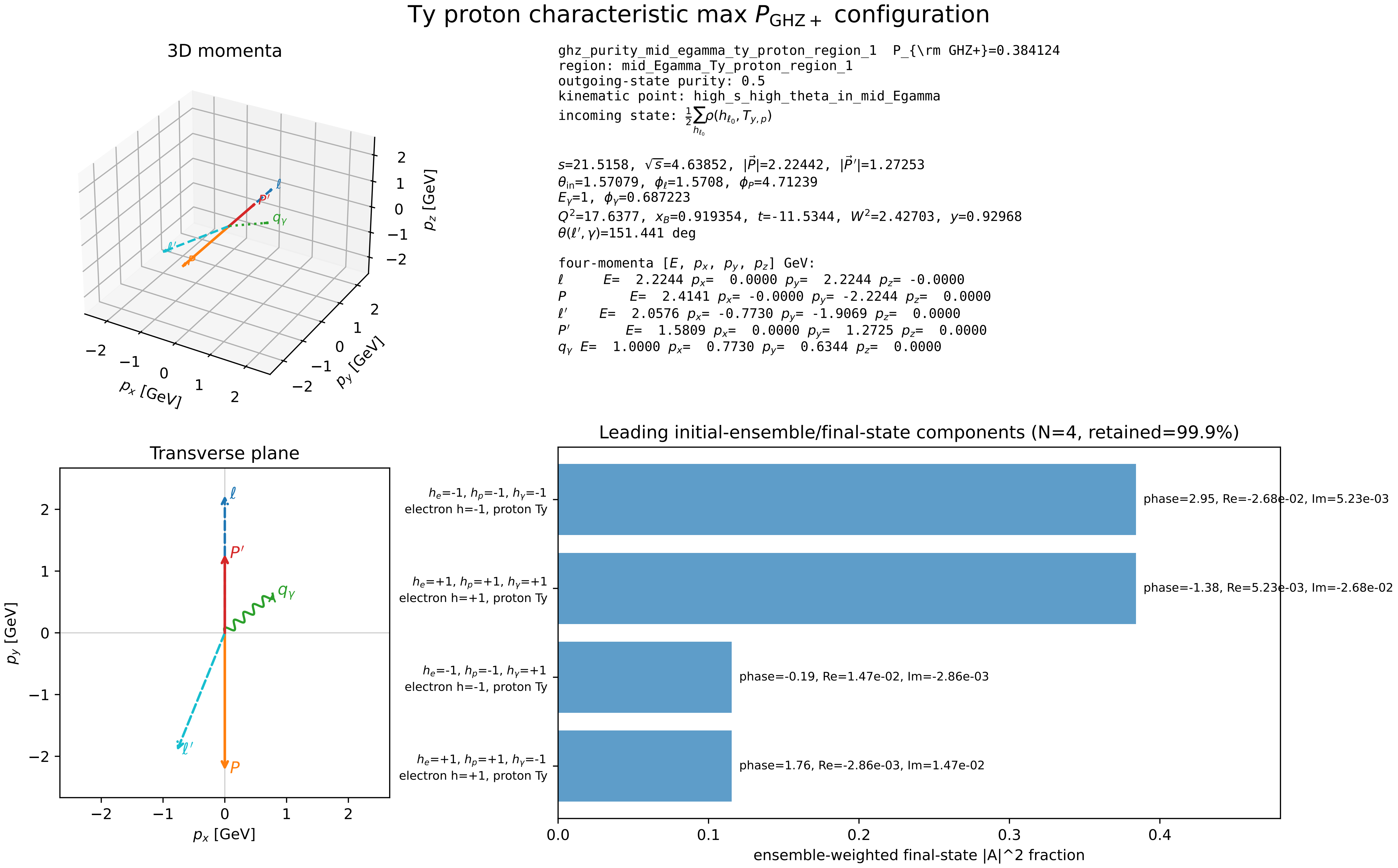
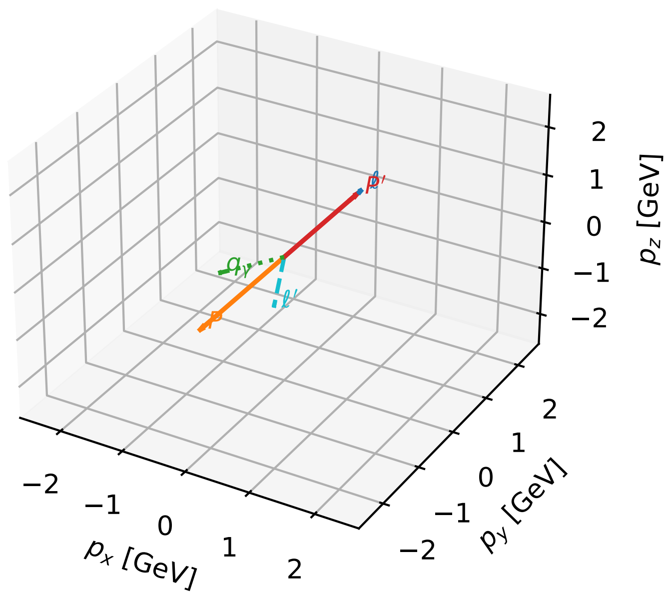

mid_Egamma Ty proton: max $P_{\mathrm{GHZ}+}$ regions
region 3 (Ty proton)
region 2 (Ty proton)
region 0 (Ty proton)
region 1 (Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Ty proton characteristic max $P_{\mathrm{GHZ+}}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
P'
q_γ
P
ℓ'
ghz_purity_mid_egamma_ty_proton_region_0 P_{\rm GHZ+}=0.384124
region: mid_Egamma_Ty_proton_region_0
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, T_{y,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.27253
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=2.45437
Q^2=17.6377, x_B=0.919354, t=-11.5344, W^2=2.42703, y=0.92968
$\theta(\ell', \gamma)$=151.441 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.0576 p_x= 0.7730 p_y= -1.9069 p_z= 0.0000
P' E= 1.5809 p_x= 0.0000 p_y= 1.2725 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7730 p_y= 0.6344 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=99.9%)
h_e=-1, h_p=-1, h_γ=-1
electron h=-1, proton Ty
phase=0.19, Re=2.68e-02, Im=5.23e-03
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton Ty
phase=1.38, Re=5.23e-03, Im=2.68e-02
h_e=-1, h_p=-1, h_γ=+1
electron h=-1, proton Ty
phase=-2.95, Re=-1.47e-02, Im=-2.86e-03
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton Ty
phase=-1.76, Re=-2.86e-03, Im=-1.47e-02
ensemble-weighted final-state |A|^2 fraction


Ty proton characteristic max $P_{\mathrm{GHZ}+}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ghz_purity_mid_egamma_ty_proton_region_1 P_{\rm GHZ+}=0.384124
region: mid_Egamma_Ty_proton_region_1
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, T_{y,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.27253
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=0.687223
Q^2=17.6377, x_B=0.919354, t=-11.5344, W^2=2.42703, y=0.92968
$\theta(\ell', \gamma)$=151.441 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.0576 p_x= -0.7730 p_y= -1.9069 p_z= 0.0000
P' E= 1.5809 p_x= 0.0000 p_y= 1.2725 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.7730 p_y= 0.6344 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=99.9%)
h_e=-1, h_p=-1, h_γ=-1
electron h=-1, proton Ty
phase=2.95, Re=-2.68e-02, Im=5.23e-03
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton Ty
phase=-1.38, Re=5.23e-03, Im=-2.68e-02
h_e=-1, h_p=-1, h_γ=+1
electron h=-1, proton Ty
phase=-0.19, Re=1.47e-02, Im=-2.86e-03
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton Ty
phase=1.76, Re=-2.86e-03, Im=1.47e-02
ensemble-weighted final-state |A|^2 fraction

Ty proton characteristic max $P_{\mathrm{GHZ}+}$ configuration

3D momenta



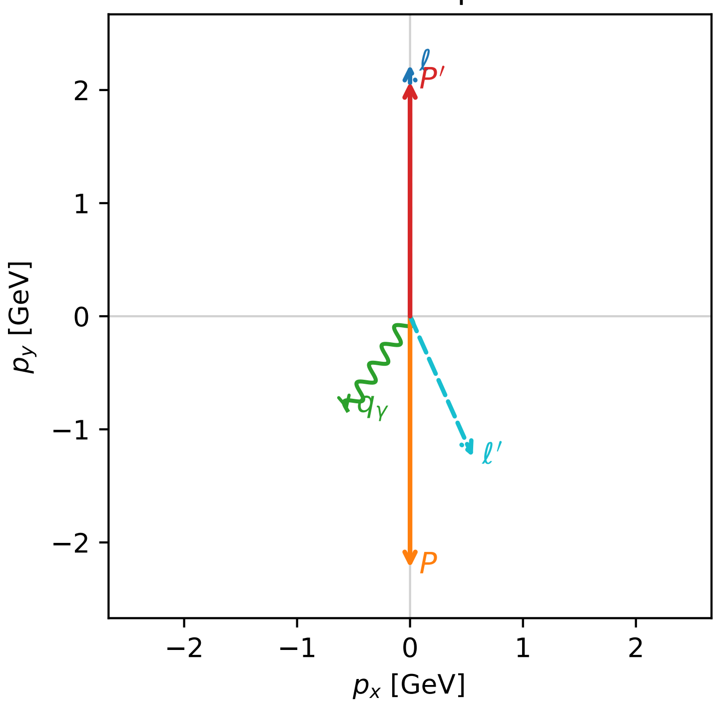
ghz_purity_mid_egamma_ty_proton_region_2 P_{\rm GHZ+}=0.383886
region: mid_Egamma_Ty_proton_region_2
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, T_{y,p})$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.07465
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=4.12334
Q^2=11.5886, x_B=0.591487, t=-18.4631, W^2=8.88355, y=0.949422
$\theta(\ell', \gamma)$=57.8296 deg

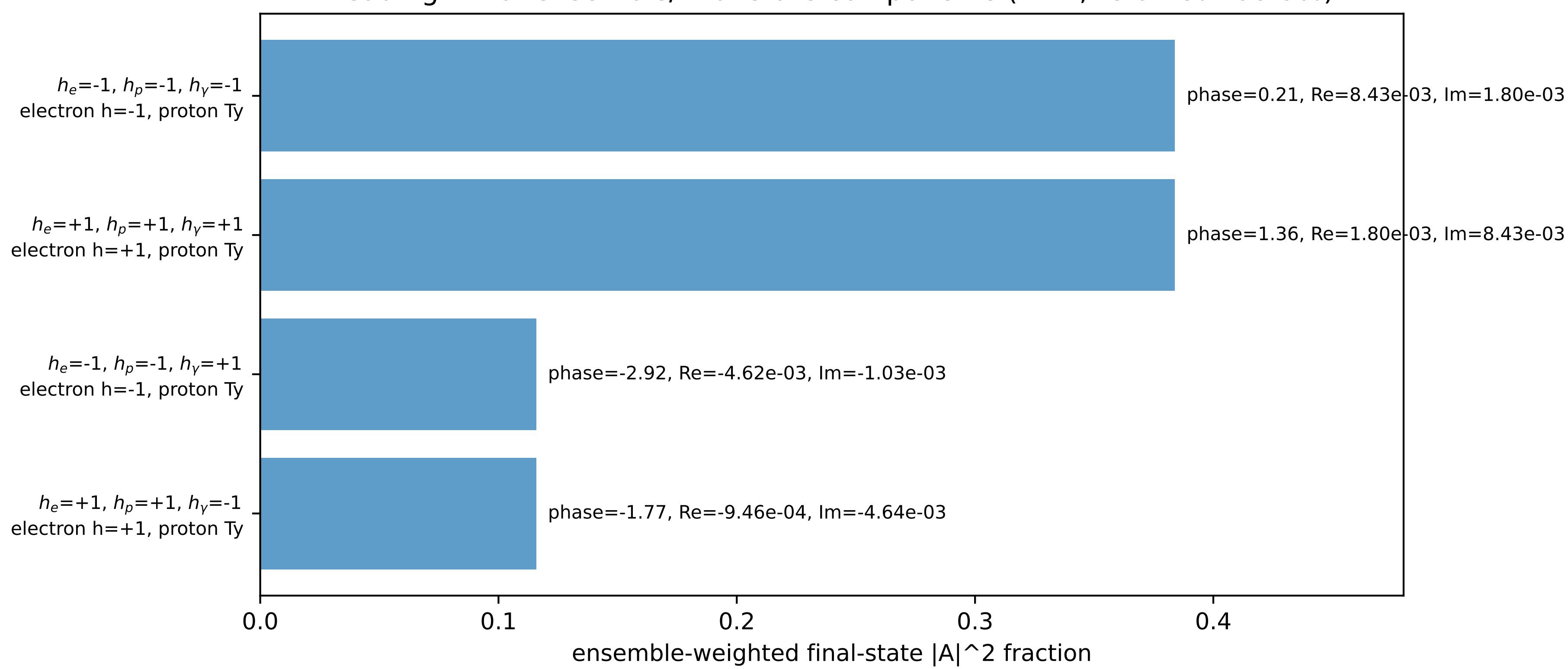
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= 0.0000	p_y= 2.2244	p_z= -0.0000
P	E= 2.4141	p_x= -0.0000	p_y= -2.2244	p_z= 0.0000
ℓ'	E= 1.3617	p_x= 0.5556	p_y= -1.2432	p_z= 0.0000
P'	E= 2.2768	p_x= 0.0000	p_y= 2.0746	p_z= 0.0000
q_γ	E= 1.0000	p_x= -0.5556	p_y= -0.8315	p_z= 0.0000

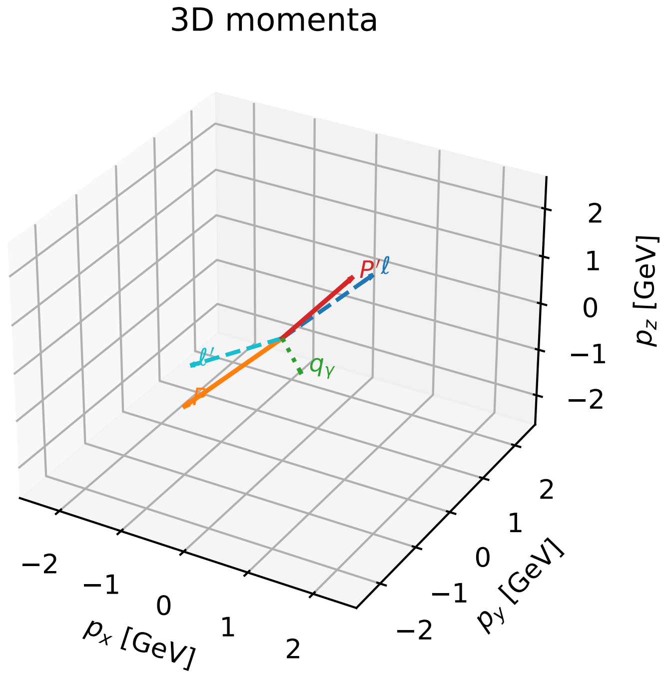
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)



Ty proton characteristic max $P_{\rm GHZ+}$ configuration



ghz_purity_mid_egamma_ty_proton_region_3 P_{\rm GHZ+}=0.383831
region: mid_Egamma_Ty_proton_region_3
outgoing-state purity: 0.500002
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, T_{y,p})$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.9752
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1, ϕ_γ=5.49779
Q^2=12.382, x_B=0.633397, t=-17.5427, W^2=8.0464, y=0.947304
$\theta(\ell', \gamma)$=74.1447 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= 0.2180	p_y= 2.2137	p_z= -0.0000
P	E= 2.4141	p_x= -0.2180	p_y= -2.2137	p_z= 0.0000
ℓ'	E= 1.4519	p_x= -0.7071	p_y= -1.2681	p_z= 0.0000
P'	E= 2.1866	p_x= 0.0000	p_y= 1.9752	p_z= 0.0000
q_γ	E= 1.0000	p_x= 0.7071	p_y= -0.7071	p_z= 0.0000